

Homework: 4

1

- ① Numerical Integration: Assuming a C^1 Euler-Bernoulli Beam Element (with Hermitian shape functions) and the shown nodal displacements. Assume a nonlinear moment-curvature constitutive relation for the beam as shown. Use numerical integration rules below and

Step 1: Find the element nodal force vector using $P = \int_{-1}^1 B^T M(K) d\xi$.

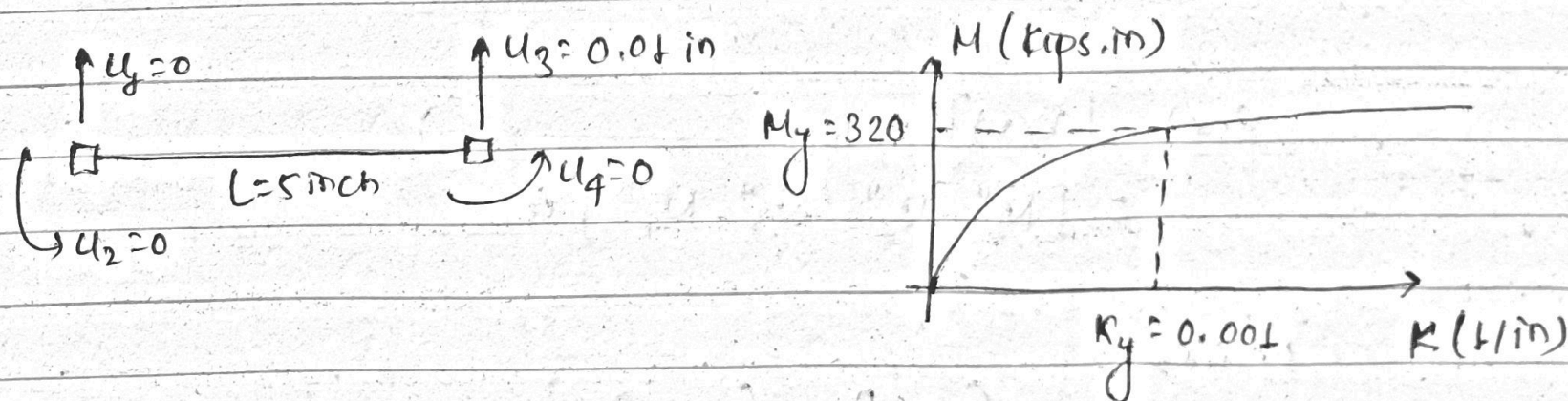
Draw the BMD.

Step 2: Find the 4×4 element tangent stiffness matrix using

$$K_T = \int_{-1}^1 B^T \frac{dM}{dK} B d\xi.$$

Compare the results between the different cases and state your observations with reason.

- ① Gaussian quadrature with 3 and 5 integration points.
- ② Gauss-Lobatto with 3, 5, and 7 integration points.



→ soln.

for C^1 Euler-Bernoulli Beam,

$$y(x) = N_1 u_1 + N_2 u_2 + N_3 u_3 + N_4 u_4 \quad \text{--- ①}$$

where, N_1, N_2, N_3 , and N_4 are Hermite shape functions given by,

$$N_1 = 1 - \frac{3x^2}{l^2} + \frac{2x^3}{l^3}$$

$$N_1'' = \frac{-6}{l^2} + \frac{12x}{l^3}$$

$$N_2 = x - \frac{2x^2}{l} + \frac{x^3}{l^2}$$

$$N_2'' = \frac{-4}{l} + \frac{6x}{l^2}$$

$$N_3 = \frac{3x^2}{l^2} - \frac{2x^3}{l^3}$$

$$N_3'' = \frac{6}{l^2} - \frac{12x}{l^3}$$

$$N_4 = -\frac{x^2}{l} + \frac{x^3}{l^2}$$

$$N_4'' = \frac{-2}{l} + \frac{6x}{l^2}$$

Compatibility:

$$\phi(x) = \frac{dy}{dx} = N_1' u_1 + N_2' u_2 + N_3' u_3 + N_4' u_4 \quad \text{--- (2)}$$

$$K(x) = \frac{d\phi}{dx} = N_1'' u_1 + N_2'' u_2 + N_3'' u_3 + N_4'' u_4 \quad \text{--- (3)}$$

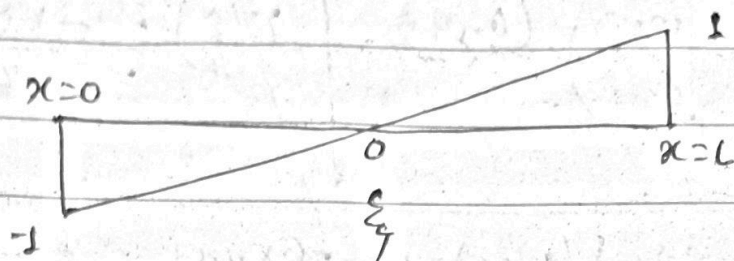
$$E(x) = [K(x)] = Bu \quad \text{--- (4)}$$

$$= [N_1'' \ N_2'' \ N_3'' \ N_4''] u$$

where,

$$u = \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0.01 \\ 0 \end{Bmatrix}$$

using natural coordinate system, $\xi \in [-1, 1]$,



$$\xi = \frac{2x}{l} - 1 \quad \Rightarrow \quad x = \frac{l}{2} (\xi + 1) \quad \left| \quad \frac{dx}{d\xi} = \frac{l}{2} = J \right.$$

$$\Rightarrow N_1'' = \frac{-6}{l^2} + \frac{12}{l^3} \left(\frac{l}{2} (\xi + 1) \right) = \frac{-6}{l^2} + \frac{6}{l^2} (\xi + 1) = \frac{6\xi}{l^2}$$

$$N_2'' = \frac{-4}{l} + \frac{6}{l^2} \cdot \frac{l}{2} (\xi + 1) = \frac{-4}{l} + \frac{3}{l} (\xi + 1) = \frac{3\xi - 1}{l}$$

$$N_3'' = \frac{6}{l^2} - \frac{12}{l^3} \cdot \frac{l}{2} (\xi + 1) = \frac{6}{l^2} - \frac{6}{l^2} (\xi + 1) = \frac{-6\xi}{l^2}$$

$$N_4'' = \frac{-2}{l} + \frac{6}{l^2} \cdot \frac{l}{2} (\xi + 1) = \frac{-2}{l} + \frac{3}{l} (\xi + 1) = \frac{3\xi + 1}{l}$$

Thus eqn (4) becomes,

$$E(\xi) = \begin{bmatrix} \frac{6\xi}{l^2} & \frac{3\xi - 1}{l} & -\frac{6\xi}{l^2} & \frac{3\xi + 1}{l} \end{bmatrix} u$$

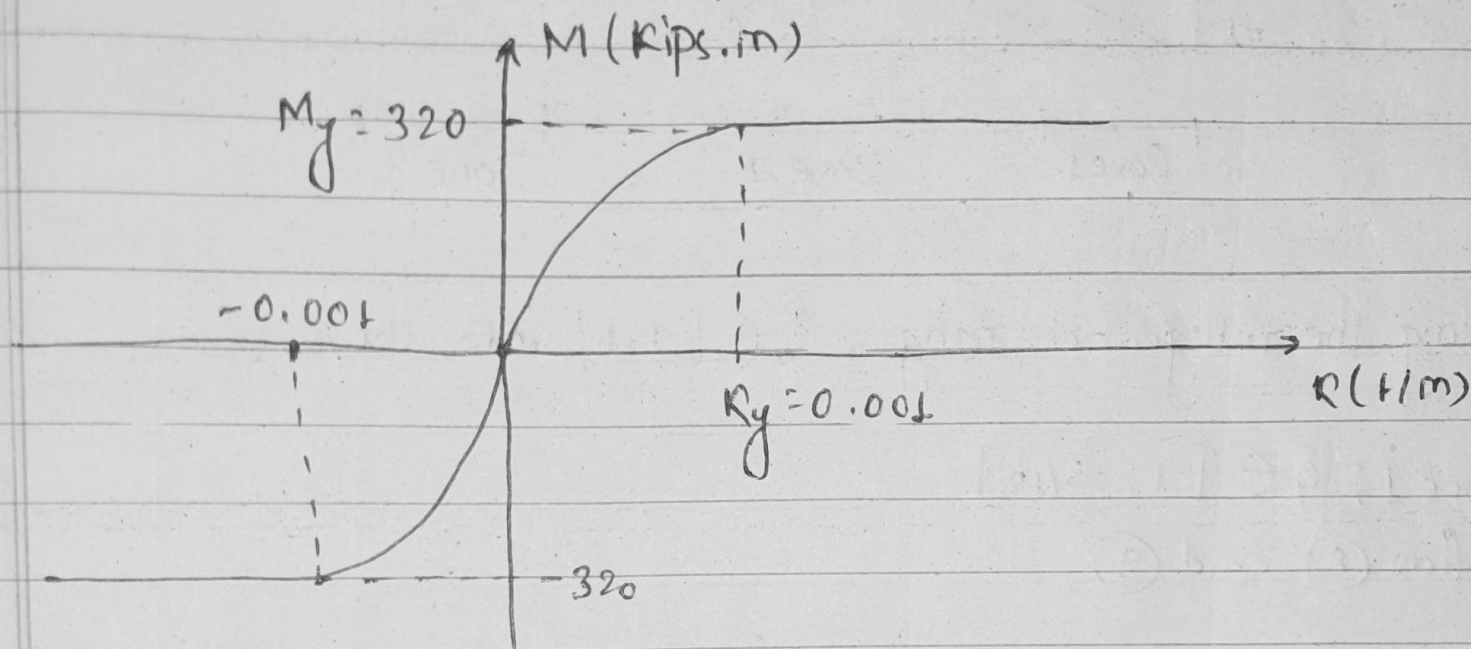
$$\Rightarrow B(\xi) = \begin{bmatrix} \frac{6\xi}{l^2} & \frac{3\xi - 1}{l} & -\frac{6\xi}{l^2} & \frac{3\xi + 1}{l} \end{bmatrix} \quad \text{--- (5)}$$

$$= \begin{bmatrix} 0.24\xi & 0.6\xi - 0.2 & -0.24\xi & 0.6\xi + 0.2 \end{bmatrix}$$

Constitutive:

$$\kappa(\xi) = \underline{B(\xi)} \underline{u} = B_3(\xi) u_3 = (-0.24\xi)(0.01) = -0.0024\xi \quad \text{--- (7)}$$

Moment - Curvature Relationship,



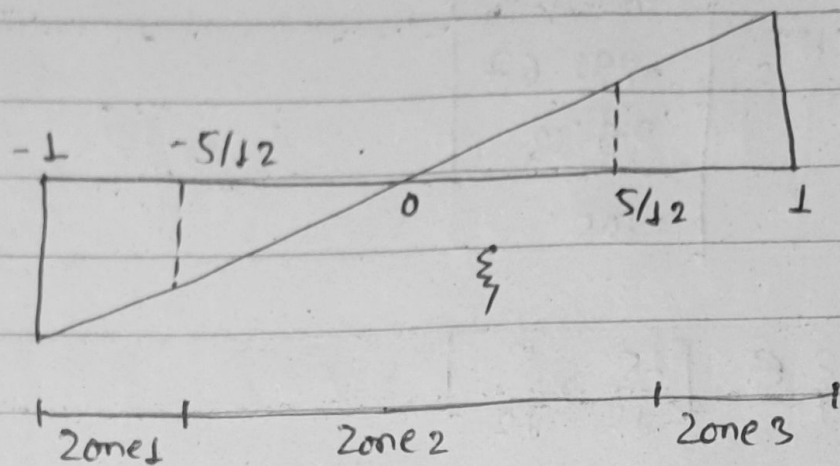
$$M(\kappa) = \begin{cases} \text{sign}(\kappa) \cdot 320 \sqrt{2000|\kappa| - 1000,000 \kappa^2} & \text{if } |\kappa| \leq 0.001 \\ \text{sign}(\kappa) \cdot 320 & \text{if } |\kappa| > 0.001 \end{cases} \quad \text{--- (8)}$$

$$\frac{dM(\kappa)}{d\kappa} = \begin{cases} \frac{320(1000 - 1000,000|\kappa|)}{\sqrt{2000|\kappa| - 1000,000 \kappa^2}} & \text{if } 0 < |\kappa| \leq 0.001 \\ 0 & \text{if } |\kappa| > 0.001 \end{cases} \quad \text{--- (9)}$$

The points where, $\kappa(\xi) \geq \kappa_y$,

$$|-0.0024\xi| \geq 0.001$$

$$\Rightarrow |\xi| \geq 5/12 \quad \text{--- (6)}$$



Dividing the integration range, $\xi \in [-1, 1]$ into three zones,

① Zone 1: $\xi \in [-1, -5/12]$

from (8) and (9),

$$M(\kappa) = 320$$

$$\frac{dM}{d\kappa} = 0$$

② Stiffness Contribution, $K_T^{(1)}$

$$K_T^{(1)} = \frac{1}{2} \int_{-1}^{-5/12} B^T \frac{dM}{d\kappa} B d\xi = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

③ Force Vector Contribution, $P^{(1)}$, $P = \int_0^1 B^T M dx = \frac{1}{2} \int_{-1}^1 B^T M d\xi$

$$P^{(1)} = \frac{5}{2} \int_{-1}^{-5/12} \begin{bmatrix} 0.24\xi \\ 0.6\xi - 0.2 \\ -0.24\xi \\ 0.6\xi + 0.2 \end{bmatrix} 320 d\xi = 800 \int_{-1}^{-5/12} \begin{bmatrix} 0.24\xi \\ 0.6\xi - 0.2 \\ -0.24\xi \\ 0.6\xi + 0.2 \end{bmatrix} d\xi$$

$$\Rightarrow P^{(1)} = \begin{bmatrix} -79.33 \\ -291.67 \\ 79.33 \\ -105 \end{bmatrix}$$

$$\textcircled{ii} \text{ Zone 2: } \xi \in \left[-\frac{5}{12}, \frac{5}{12} \right]$$

from (8) and (9),

$$M = \text{Sign}(K) 320 \sqrt{2000|K| - 1000,000K^2}$$

$$\frac{dM}{dK} = \frac{320 (1000 - 1000,000|K|)}{\sqrt{2000|K| - 1000,000K^2}}$$

(a) Stiffness Contribution, $K_T^{(2)}$,

$$K_T^{(2)} = \frac{1}{2} \int_{-5/12}^{5/12} B^T \frac{dM}{dK} B d\xi = \frac{1}{2} \int_{-5/12}^0 B^T \frac{dM}{dK} B d\xi + \frac{1}{2} \int_0^{5/12} B^T \frac{dM}{dK} B d\xi$$

$$K = -0.0024 \xi,$$

$$K_T^{(2)} = 2.5 \int_{-5/12}^{5/12} \begin{bmatrix} 0.24\xi \\ 0.6\xi - 0.2 \\ -0.24\xi \\ 0.6\xi + 0.2 \end{bmatrix} \frac{320 (1000 - 2400|\xi|)}{\sqrt{4.8|\xi| - 5.76\xi^2}} B d\xi$$

Using python,

$$K_T^{(2)} = \begin{bmatrix} 639.14 & 1597.84 & -639.14 & 1597.84 \\ 1597.84 & 30661.26 & -1597.84 & -22672.07 \\ -639.14 & -1597.84 & 639.14 & -1597.84 \\ 1597.84 & -22672.07 & -1597.84 & 30661.26 \end{bmatrix}$$

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$$P^{(2)} = 2.5 \int_{-5/12}^{5/12} B^T M d\xi$$

$$= 2.5 \int_{-5/12}^{5/12} \begin{bmatrix} 0.24\xi \\ 0.6\xi - 0.2 \\ -0.24\xi \\ 0.6\xi + 0.2 \end{bmatrix} 320 \sqrt{2000|K|} = 1000,000 K^2 d\xi$$

$$P^{(2)} = \begin{bmatrix} -30.14 \\ -75.34 \\ 30.14 \\ -75.34 \end{bmatrix}$$

III) Zone 3: $\xi \in [5/12, 1]$

from (8) and (9),

$$M = -320$$

$$\frac{dM}{dK} = 0$$

$$\therefore K_T^{(3)} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

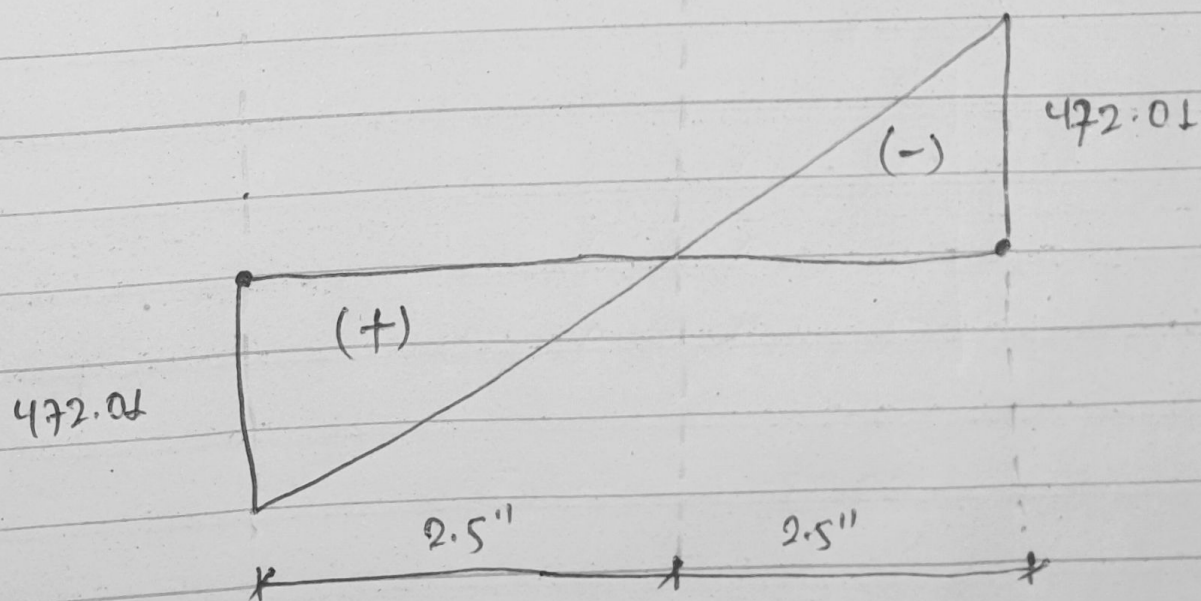
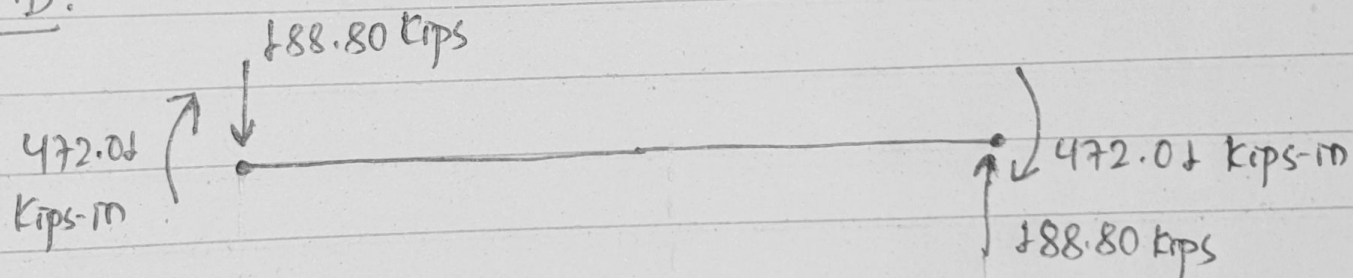
$$P^{(3)} = -800 \int_{5/12}^1 B^T d\xi = \begin{bmatrix} -79.33 \\ -105 \\ 79.33 \\ -291.67 \end{bmatrix}$$

Analytical Element Tangent Stiffness Matrix:

$$K_T = \begin{bmatrix} 639.14 & 1597.84 & -639.14 & 1597.84 \\ 1597.84 & 30661.26 & -1597.84 & -22672.07 \\ -639.14 & -1597.84 & 639.14 & -1597.84 \\ 1597.84 & -22672.07 & -1597.84 & 30661.26 \end{bmatrix}$$

$$P = \begin{bmatrix} -188.80 \\ -472.01 \\ 188.80 \\ -472.01 \end{bmatrix} = P^{(1)} + P^{(2)} + P^{(3)}$$

BMD:



Problem # 1

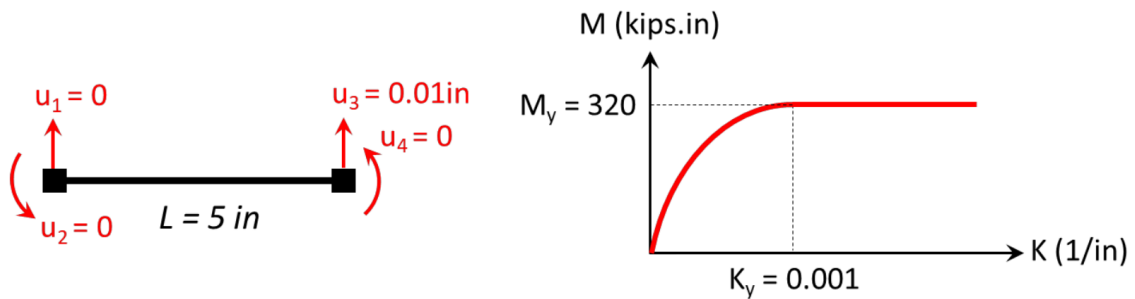
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Step 1: Find the element nodal force vector using $\mathbf{P} = \int_{-1}^1 \mathbf{B}^T \mathbf{M}(\kappa) d\xi$. Draw the BMD.

Step 2: Find the 4x4 element tangent stiffness matrix using $\mathbf{K}_T = \int_{-1}^1 \mathbf{B}^T \frac{d\mathbf{M}}{d\kappa} \mathbf{B} d\xi$.

Compare the results between the different cases and state your observations with reason.

- a. Gaussian quadrature with 3 and 5 integration points.
- b. Gauss-Lobatto with 3, 5, and 7 integration points.



Codes:

<https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Curve.py>

https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Q1_revised.py

https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Q1_integration.py

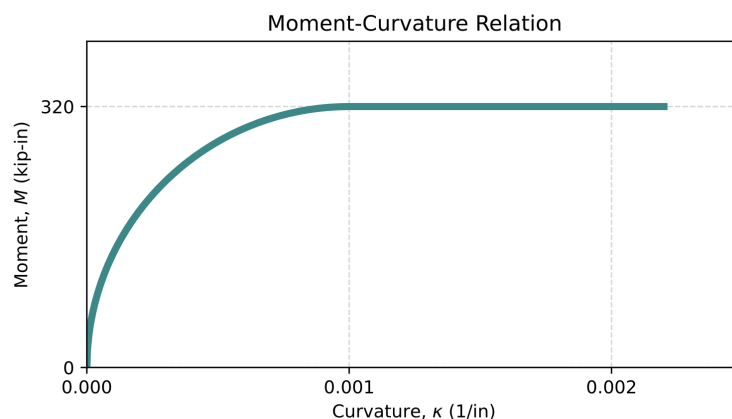


Fig. Adopted Moment Curvature Relationship

Moment-Curvature Relationship $M(\kappa)$

$$M(\kappa) = \begin{cases} \text{sign}(\kappa) \cdot 320 \sqrt{2000|\kappa| - 1,000,000\kappa^2} & \text{if } |\kappa| \leq 0.001 \\ \text{sign}(\kappa) \cdot 320 & \text{if } |\kappa| > 0.001 \end{cases}$$

Tangent Modulus $\frac{dM}{d\kappa}$

$$\frac{dM}{d\kappa} = \begin{cases} \frac{320(1000 - 1,000,000|\kappa|)}{\sqrt{2000|\kappa| - 1,000,000\kappa^2}} & \text{if } 0 < |\kappa| \leq 0.001 \\ 0 & \text{if } |\kappa| > 0.001 \end{cases}$$

Analytical Stiffness Matrix (K_T),

$$\begin{bmatrix} 639.14 & 1597.84 & -639.14 & 1597.84 \\ 1597.84 & 30661.26 & -1597.84 & -22672.07 \\ -639.14 & -1597.84 & 639.14 & -1597.84 \\ 1597.84 & -22672.07 & -1597.84 & 30661.26 \end{bmatrix}$$

Analytical Nodal Force Matrix (P) = $[P_1, P_2, P_3, P_4]^T$

$$\begin{bmatrix} -188.80 \\ -472.01 \\ 188.80 \\ -472.01 \end{bmatrix}$$

Gauss quadrature (3 Points): $\xi = [-0.7746, 0.0, 0.7746]$

Nodal Force Vector P:

$$\begin{bmatrix} -165.25 \\ -413.12 \\ 165.25 \\ -413.12 \end{bmatrix}$$

Tangent Stiffness Matrix K_T :

$$\begin{bmatrix} 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 28444.44 & 0.00 & -28444.44 \\ 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & -28444.44 & 0.00 & 28444.44 \end{bmatrix}$$

Gauss quadrature (5 Points): $\xi = [-0.9062, -0.5384, 0.0, 0.5384, 0.9062]$

Nodal Force Vector P:

[[-181.41]
[-453.53]
[181.41]
[-453.53]]

Tangent Stiffness Matrix K_T :

[[0.00 0.00 0.00 0.00]
[0.00 18204.44 0.00 -18204.44]
[0.00 0.00 0.00 0.00]
[0.00 -18204.44 0.00 18204.44]]

Gauss lobatto (3 Points): $\xi = [-1.0, 0.0, 1.0]$

Nodal Force Vector P:

[[-128.00]
[-320.00]
[128.00]
[-320.00]]

Tangent Stiffness Matrix K_T :

[[0.00 0.00 0.00 0.00]
[0.00 42666.67 0.00 -42666.67]
[0.00 0.00 0.00 0.00]
[0.00 -42666.67 0.00 42666.67]]

Gauss lobatto (5 Points): $\xi = [-1.0, -0.6547, 0.0, 0.6547, 1.0]$

Nodal Force Vector P:

[[-175.27]
[-438.17]
[175.27]
[-438.17]]

Tangent Stiffness Matrix K_T :

[[0.00 0.00 0.00 0.00]
[0.00 22755.56 0.00 -22755.56]
[0.00 0.00 0.00 0.00]
[0.00 -22755.56 0.00 22755.56]]

Gauss lobatto (7 Points): $\xi = [-1.0, -0.8302, -0.4688, 0.0, 0.4688, 0.8302, 1.0]$

Nodal Force Vector P:

```
[[ -184.27]
 [ -460.68]
 [  184.27]
 [ -460.68]]
```

Tangent Stiffness Matrix K_T :

```
[[ 0.00    0.00    0.00    0.00]
 [ 0.00  15603.81  0.00 -15603.81]
 [ 0.00    0.00    0.00    0.00]
 [ 0.00 -15603.81  0.00  15603.81]]
```

Observations

1. Nodal Force Vector (P) Comparison: As the number of integration points increases for both Gaussian quadrature and Gauss-Lobatto schemes, the calculated nodal force vector P converges toward the analytical solution. This improvement occurs because a greater number of integration points enables the numerical integration procedure to sample the integrand at more locations, while the associated weighting factors provide a more accurate approximation of the exact integral.

2. Tangent Stiffness Matrix (K_T): All five tested numerical schemes yield an incomplete tangent stiffness matrix where rows and columns 1 and 3 are entirely zero. The nonlinear core of the beam is localized within $\xi = [-0.4167, 0.4167]$. In all 3, 5, and 7-point rules, the only integration point that lands inside this zone is $\xi = 0$. Thus, K_T is driven exclusively by the single sample point at $\xi = 0$. At $\xi = 0$, B Matrix is,

```
[[ 0]
 [ -0.2]
 [ 0]
 [ 0.2]]
```

Because the translational components (indices 1 and 3) are exactly 0 at the center of the beam, the outer product $B(0)^T \cdot B(0)$ results in a matrix where rows/columns 1 and 3 are entirely zeroed out. Since no other integration points are inside this zone to provide translational stiffness contributions, the final accumulated numerical matrix results in those zeros.

3. Fluctuations in Rotational Stiffness: The rotational stiffness values (indices 2 and 4) fluctuate drastically between schemes rather than converging smoothly. Since $\xi = 0$ is the only point evaluating a non-zero material stiffness ($dM/d\kappa > 0$), the entire matrix is completely dependent to the mathematical weight assigned at the center point by each specific integration scheme.

To capture the true matrix and get rid of the zero rows/columns, we should use an integration rule high enough to place points between 0 and 0.4167.

- ② Calculate the analytical material tangent modulus, $E_t = \frac{d\sigma}{d\epsilon}$ for the Menegotto-Pinto material model presented in the class, see uploaded "Menegotto-Pinto Uniaxial Material Model, pdf". Check your derived tangent modulus with the programmed one in the Bauschinger function of the Matlab Code.

→ Solⁿ:

For Menegotto-Pinto Model we have,

$$\sigma^* = b\epsilon^* + \frac{(1-b)\epsilon^*}{(1+|\epsilon^*|^R)^{1/R}} \quad \text{--- (1)}$$

where,

$$\sigma^* = \text{normalized stress} = \frac{\sigma - \sigma_r}{\sigma_y - \sigma_r} \quad \text{--- (2)}$$

$$\epsilon^* = \text{normalized strain} = \frac{\epsilon - \epsilon_r}{\epsilon_y - \epsilon_r} \quad \text{--- (3)}$$

(ϵ_r, σ_r) is the last strain reversal (unloading) point,
 b is the strain hardening ratio,
 R controls the curvature of elastic to plastic transition

Now,

$$E_t = \frac{d\sigma}{d\epsilon} = \frac{d\sigma}{d\sigma^*} \frac{d\sigma^*}{d\epsilon^*} \frac{d\epsilon^*}{d\epsilon} \quad \text{--- (4)}$$

from equation (3),

$$\frac{d\epsilon^*}{d\epsilon} = \frac{1}{\epsilon_y - \epsilon_r} \quad \text{--- (a)}$$

from equation (2),

$$\frac{d\phi^*}{d\phi} = \frac{1}{\phi_y - \phi_r} \Rightarrow \frac{d\phi}{d\phi^*} = \phi_y - \phi_r \quad (b)$$

And,

$$\begin{aligned} \frac{d}{d\epsilon^*} \left\{ \frac{\epsilon^*}{(1 + |\epsilon^*|^R)^{1/R}} \right\} &= \frac{d}{d\epsilon^*} \left\{ \epsilon^* \cdot (1 + |\epsilon^*|^R)^{-1/R} \right\} \\ &= (1 + |\epsilon^*|^R)^{-1/R} + \epsilon^* \left(-\frac{1}{R} \right) (1 + |\epsilon^*|^R)^{-\frac{1}{R}-1} (R) |\epsilon^*|^{R-1} \text{sign}(\epsilon^*) \\ &= (1 + |\epsilon^*|^R)^{-1/R} - \left\{ \epsilon^* |\epsilon^*|^{R-1} \text{sign}(\epsilon^*) \right\} (1 + |\epsilon^*|^R)^{-\frac{1}{R}-1} \\ &= (1 + |\epsilon^*|^R)^{-\frac{1}{R}} - |\epsilon^*|^R (1 + |\epsilon^*|^R)^{-\frac{1}{R}-1} \\ &= (1 + |\epsilon^*|^R)^{-\frac{1}{R}-1} \left\{ (1 + |\epsilon^*|^R) - |\epsilon^*|^R \right\} \\ &= (1 + |\epsilon^*|^R)^{-\frac{1}{R}-1} \quad (c) \end{aligned}$$

from equation (1),

$$\frac{d\phi^*}{d\epsilon^*} = b + (1-b) \frac{d}{d\epsilon^*} \left\{ \frac{\epsilon^*}{(1 + |\epsilon^*|^R)^{1/R}} \right\}$$

$$= b + (1-b) \cdot \frac{1}{(1+|E^*|^R)^{\frac{1}{R}+1}} \quad \text{--- (d)}$$

Putting the values from (a), (b), (d) in equation (4),

$$E_T = \frac{dG}{dE} = \left(\frac{G_y - G_r}{E_y - E_r} \right) \left\{ b + (1-b) \cdot \frac{1}{(1+|E^*|^R)^{\frac{1}{R}+1}} \right\} \quad \text{--- (5)}$$

from the Bauschinger function of the Matlab code,

$$\begin{aligned} dsdE &= b + (1-b) \times \frac{\left(1 - \frac{|E^*|^R}{1+|E^*|^R} \right)}{\left(1 + |E^*|^R \right)^{\frac{1}{R}}} \\ &= b + (1-b) \cdot \frac{1}{(1+|E^*|^R)^{\frac{1}{R}+1}} \end{aligned}$$

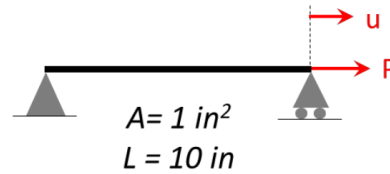
And, Here $G_0 = G_y$,

$$E_T = \left(\frac{G_0 - G_r}{E_0 - E_r} \right) dsdE = \left(\frac{G_0 - G_r}{E_0 - E_r} \right) \left\{ b + (1-b) \cdot \frac{1}{(1+|E^*|^R)^{\frac{1}{R}+1}} \right\}$$

which is same as equation (5).

Problem # 3

3. Incremental Iterative Solution: Consider a truss element under axial loading as shown. The material is modeled using Menegotto-Pinto model with the following parameters: $\sigma_y = 60 \text{ ksi}$, $E = 29000 \text{ ksi}$, $b = 0.05$, $R_0 = 20$, $R_1 = 0.925$, $R_2 = 0.15$, $a_1 = a_2 = 0$.



Write a Matlab code to perform incremental iterative process for quasi-static loadings as detailed below. Use the Menegotto-Pinto material model code provided in the class. Return your commented Matlab code with your solution.

- Perform a quasi-static monotonic pushover analysis from $P = 0$ to $P = 200$ kips using Newton Raphson method. Plot the pushover curve (i.e., force vs. displacement). Plot the iteration points on top of the pushover curve.
Note: a good force step size is 3-5% of the yield force.
- Repeat part (a) with modified Newton algorithm using the initial elastic stiffness. Plot the convergence criterion vs. number of iterations and compare for the two methods. State your observations.
- Perform a quasi-static cyclic pushover analysis for $P = [0, 100, -100, 200, 0]$ kips. Plot the cyclic pushover curve. If convergence cannot be achieved, try to investigate the underlying reason(s) and solve the issues. State the reason(s) and the solution you developed.

Code: <https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Q3.py>

Part a and b: Observations

Newton-Raphson: The maximum number of iterations in any single load step is only 5, resulting in a total of 217 iterations across the entire pushover analysis. This method updates the global tangent stiffness at each iteration, allowing the algorithm to rapidly find equilibrium in very few iterations, even after significant yielding.

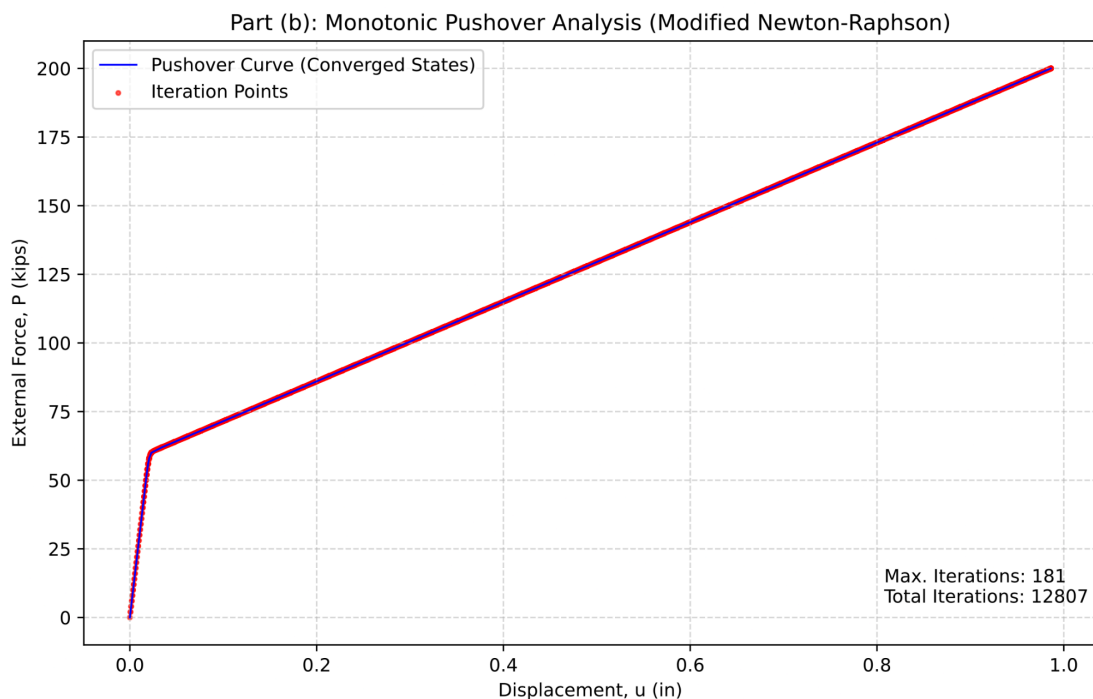
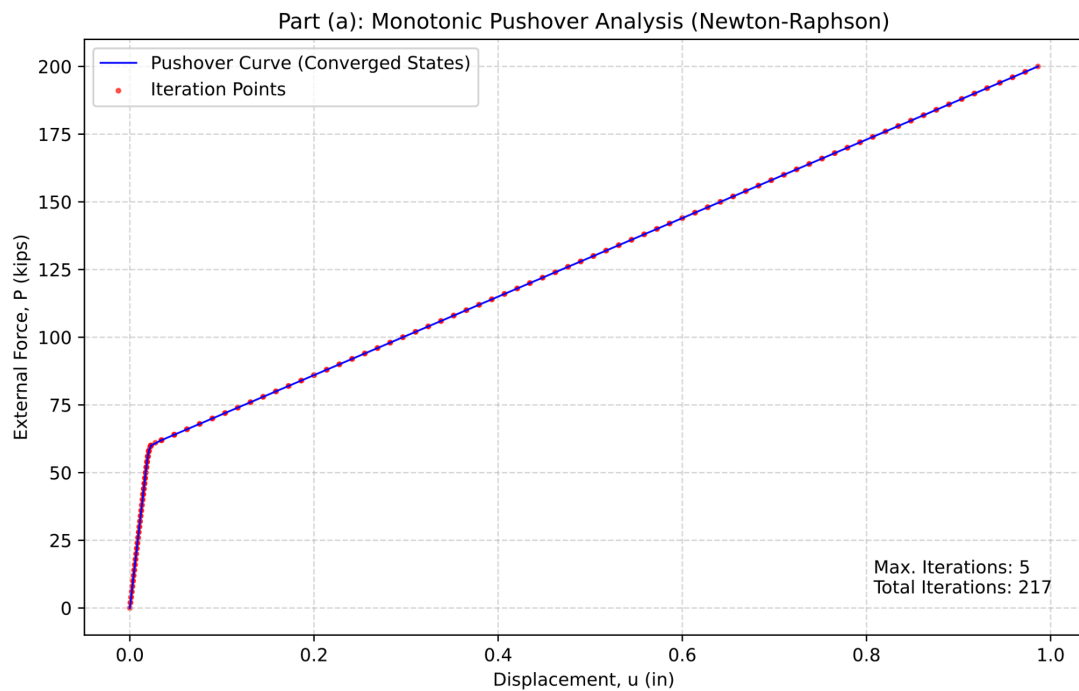
Modified Newton-Raphson: The maximum number of iterations in any single load step is 181, resulting in a total of 12,807 iterations across the entire pushover analysis to complete the identical load history. This method uses the same initial global tangent stiffness throughout the analysis.

Pre-Yield: The structural response is perfectly elastic, so K_{initial} matches the true tangent stiffness, and convergence is achieved in very few steps using both algorithms.

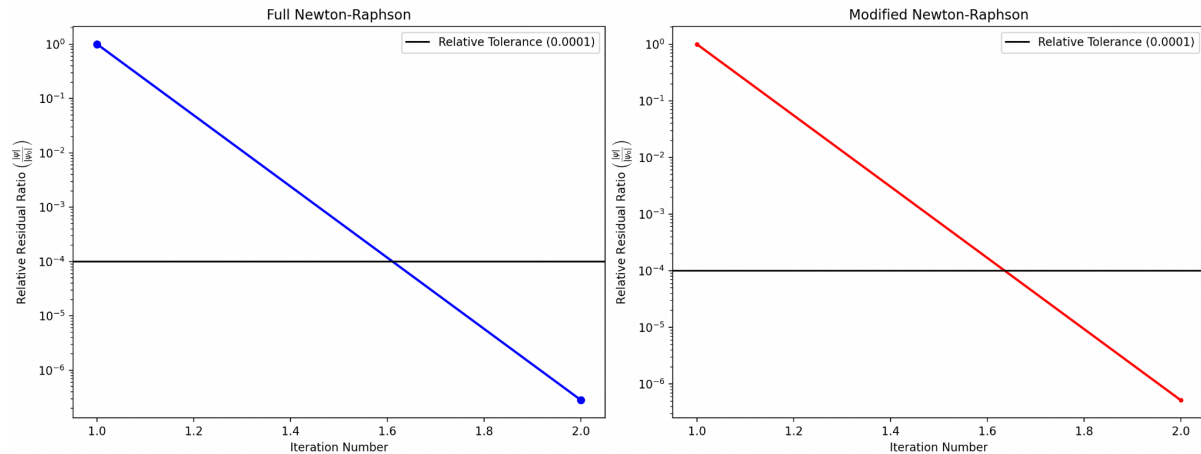
Post-Yield: The member yields, and the true tangent stiffness drops to just 5% of its elastic value (as $b=0.05$). Since it continues to use K_{initial} , it under predicts the necessary displacement correction in each iteration. This leads to a very slow convergence rate.

Convergence Criterion Adopted: Relative Norm Unbalanced Force, $\varepsilon = 10^{-4}$

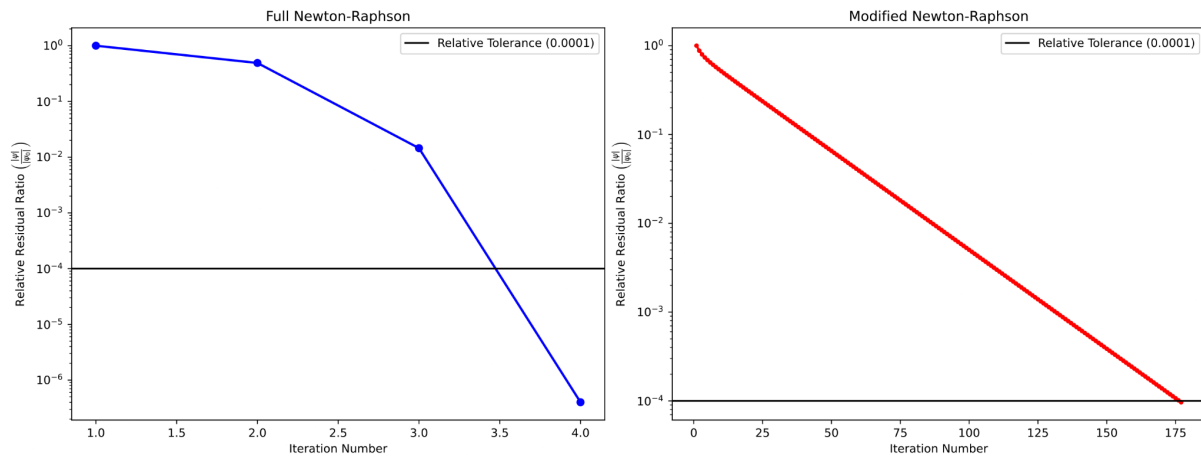
Convergence Criterion: Just after the yield point, the relative residual force ratio is plotted side by side for Full Newton-Raphson and Modified Newton-Raphson. Full Newton-Raphson shows Quadratic Convergence achieving the required tolerance in just 4 iterations. This is because the tangent stiffness updates at every iteration. In contrast, Modified Newton-Raphson shows Linear Convergence 175 iterations to achieve the same tolerance. This is because it keeps using the initial tangent stiffness.



Part (b): Convergence comparison before yielding



Part (b): Convergence comparison just after yield point



Part c: Observations, Underlying Reasons, and Solution:

There is convergence issue at each reversal: When the cyclic load path is run with Newton-Raphson, convergence fails at the load steps right after each reversal (each peak). The solver hits the maximum iterations and prints warnings such as:

“

Warning: Reached max iterations at load P = 98.0 using Full-NR

Warning: Reached max iterations at load P = 96.0 using Full-NR

Warning: Reached max iterations at load P = 94.0 using Full-NR

....

Warning: Reached max iterations at load P = -98.0 using Full-NR

Warning: Reached max iterations at load P = -96.0 using Full-NR

Warning: Reached max iterations at load P = -94.0 using Full-NR

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Warning: Reached max iterations at load P = 198.0 using Full-NR

Warning: Reached max iterations at load P = 196.0 using Full-NR

Warning: Reached max iterations at load P = 194.0 using Full-NR

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Reason: This is due to the slope discontinuity (the kink) at every reversal. At a converged peak (say $P = 100$ kips) the material is on the hardening branch, so the stored tangent modulus is $E_t \approx bE = 0.05 \times 29000 = 1450$ ksi, and the structural tangent stiffness is $K_T = A \cdot E_t / L = 145$ kip/in, which is small. The next load step asks for slightly less force, but the displacement correction $\delta u = \psi / K_T$ is large. This large correction moves the displacement far past the equilibrium point, across the kink and onto the steep elastic-unloading branch. On the next iteration it overshoots back. The trial strain increment keeps flipping sign, so the material keeps jumping between the loading and unloading branches and never settles within the tolerance in the allowed number of iterations.

Solution: To solve this issue, for the first iteration of every load step we use elastic stiffness, and the consistent tangent for all later iterations:

$$K_T = K_{\text{elastic}} \quad \text{if } \text{idx} == 0 \quad \text{else} \quad (A * \text{state}.E_t) / L$$

The elastic stiffness being large (2900 ksi), the first correction $\delta u = \psi / K_{\text{elastic}}$ is small and lands on the correct (unloading) branch instead of jumping across the discontinuity. After that one small step, the consistent tangent is used which converges quickly.

With this solution, all cyclic load steps converged (at most 7 iterations per step) and the correct hysteresis loop was obtained.

There were no such issues during part a and b because during monotonic loading there were no reversals.

